



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE (Advanced)

SCORE-3.0

20-06-2021

JEE(Main + Advance) : Leader & Enthusiast Course (PHASE : SCORE-3.0)

Test Type : SCORE-3.0

Test Pattern : JEE (Advanced)

TEST DATE : 20-06-2021

ANSWER KEY

PAPER-1

PHYSICS

SEC-1	Q.	1	2	3	4	5	6		
	A.	B,C,D	A,B,C	A,B,D	B,C	A,C,D	B,C,D		
SEC-2	Q.	1	2	3	4				
	A.	A	B	B	D				
SEC-3	Q.	1	2	3	4	5	6	7	8
	A.	1.00	3.36	6.94	6.28	100.00	6.28	30.00	1.57 to 1.58

CHEMISTRY

SEC-1	Q.	1	2	3	4	5	6		
	A.	B,C,D	A,B,D	A,B,C,D	A,B,C	B,C	C		
SEC-2	Q.	1	2	3	4				
	A.	A	D	C	B				
SEC-3	Q.	1	2	3	4	5	6	7	8
	A.	0.75	4.07 to 4.08	2.50	6.00	11.00	4.00	5.00	3.00

MATHEMATICS

SEC-1	Q.	1	2	3	4	5	6		
	A.	A,D	A,B,C,D	A,B,D	B,D	B	A,B,D		
SEC-2	Q.	1	2	3	4				
	A.	C	B	D	B				
SEC-3	Q.	1	2	3	4	5	6	7	8
	A.	6.28	1.00	1.00	0.75	3.14	1.50	3.00	3.00

HINT – SHEET

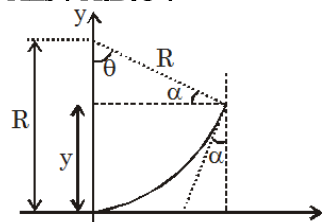
PHYSICS

SEC-1

1. Ans (B,C,D)

$$\beta = \frac{\lambda}{2\mu\theta}$$

2. Ans (A,B,C)



$$\mu_0 \sin 90^\circ = \mu \sin \alpha$$

$$\frac{\mu_0}{\sin \alpha} = \mu$$

$$\mu = \frac{\mu_0 R}{R - y}$$

$$\mu_0 = 1$$

$$\sin \alpha = \frac{R - y}{R}$$

$$y = \frac{R}{2}$$

$$\sin \alpha = \frac{1}{2}$$

$$\alpha = 30^\circ$$

unit vector is $\sin \alpha \hat{i} + \cos \alpha \hat{j}$

$$\frac{5}{2} = \mu = \frac{R}{R - y} \Rightarrow 5(R - y) = 2R \Rightarrow y = \frac{3R}{5}$$

$$\cos \theta_{\max} = \frac{2}{5}$$

5. Ans (A,C,D)

Since $0 < M < \infty$ therefore options A and D are incorrect.

Consider option B

$$\frac{\mu}{M} = \frac{2\pi}{\lambda} \tan \frac{2\pi L}{\lambda}$$

$$\text{If } M \rightarrow \infty \text{ then } \frac{\mu}{M} \rightarrow 0$$

$$\tan \frac{2\pi L}{\lambda} \rightarrow 0$$

$$\frac{2\pi L}{\lambda} \rightarrow n\pi$$

$$L \rightarrow \frac{n\lambda}{2}$$

$$\text{If } M \rightarrow 0 \text{ then } \frac{\mu}{M} \rightarrow \infty$$

$$\tan \frac{2\pi L}{\lambda} \rightarrow \infty$$

$$\frac{2\pi L}{\lambda} \rightarrow \left(n + \frac{1}{2}\right)\pi$$

$$L \rightarrow \left(n + \frac{1}{2}\right) \frac{\lambda}{4}$$

Consider option C

$$\text{If } M \rightarrow 0 \text{ then } \frac{\mu}{M} \rightarrow \infty$$

$$\sin \frac{2\pi L}{\lambda} \rightarrow \infty$$

Not possible

If $M = \infty$ then option D is correct.

But $0 < M < \infty$, hence option D is incorrect.

6. Ans (B,C,D)

Let maximum intensity be I_0

$$I = I_0 \cos^2 \left(\frac{\Delta\phi}{2} \right)$$

$$I_{av} = \frac{I_0}{2}$$

$$I < I_{av} \Rightarrow \cos^2 \left(\frac{\Delta\phi}{2} \right) < \frac{1}{2}$$

$$\Rightarrow \cos \left(\frac{\Delta\phi}{2} \right) < \frac{1}{\sqrt{2}}$$

$$\Rightarrow \frac{\pi}{2} < \Delta\phi < \frac{3\pi}{2}$$

PHYSICS

SEC-2

1. Ans (A)

$$P_{av} = \frac{1}{2} \mu v \omega^2 A^2$$

$$P' = 0.5 P_{av}$$

$$= 0.01145 \text{ Watt}$$

2. Ans (B)

$$Q = P't$$

$$mC\Delta T = P't$$

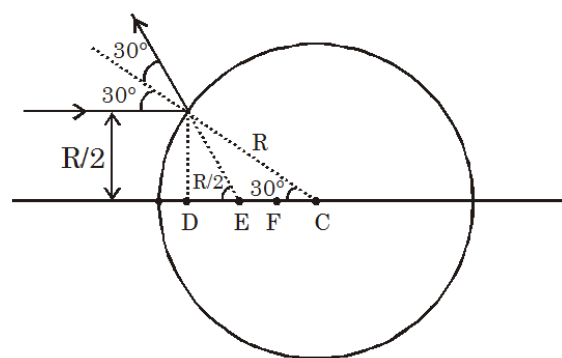
$$t = \frac{mC\Delta T}{P'}$$

$$t \simeq 4.2 \text{ days}$$

PHYSICS

SEC-3

1. Ans (1.00)



$$CD = \frac{R \cot 30^\circ}{2} = \frac{3m}{2}$$

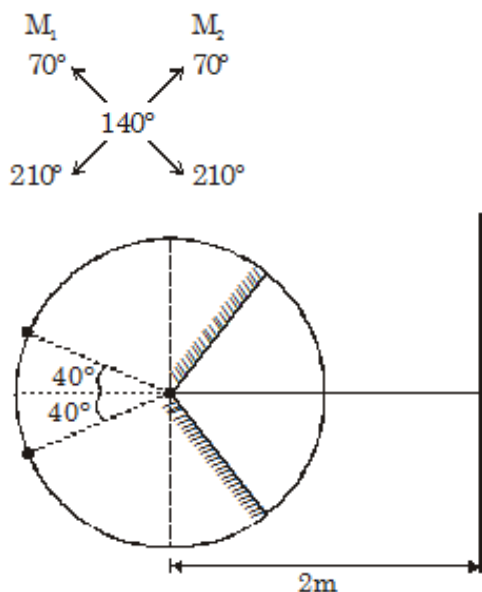
$$ED = \frac{R}{2} \cot 60^\circ$$

$$= \frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{3}} = \frac{1}{2} m$$

$$CE = (1.5 - 0.5)m$$

$$= 1m$$

2. Ans (3.36)



$$D = 2(2 \tan 40^\circ)$$

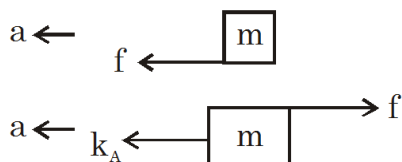
$$D = 4 \times 0.84 = 3.36 \text{ m}$$

3. Ans (6.94)

$$T = \frac{\pi}{6} = 2\pi \sqrt{\frac{2}{k}}$$

$$\left(\frac{1}{12}\right)^2 = \frac{2}{k}$$

$$k = 2(12)^2 \text{ N/m}$$



$$ma = f$$

$$kA - f = Ma$$

$$kA - f = M(f/m)$$

$$kA = \left(\frac{M}{m} + 1\right) f$$

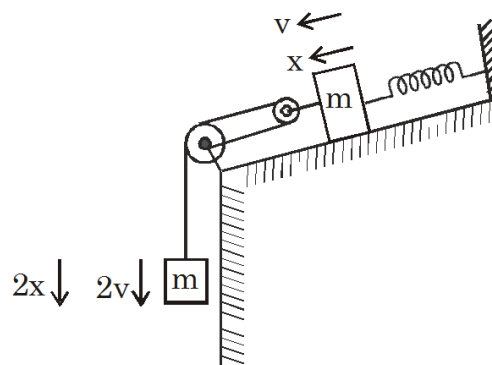
$$f = \left(\frac{kAm}{M+m}\right) \leq \mu mg$$

$$\frac{kA}{M} \leq \mu g$$

$$A \leq \frac{\mu g M}{k}$$

$$A \leq \frac{0.1 \times 10 \times 2}{(2)(12)^2}$$

4. Ans (6.28)



$$E = \frac{1}{2}mv^2 + \frac{1}{2}m(2v)^2 + \frac{1}{2}kx^2$$

$$E = 5 \left(\frac{1}{2}mv^2\right) + \frac{1}{2}kx^2$$

$$T = 2\pi \sqrt{\frac{5m}{k}}$$

$$T = 2\pi \sqrt{\frac{5 \times 1}{5}}$$

$$T = 2\pi$$

5. Ans (100.00)

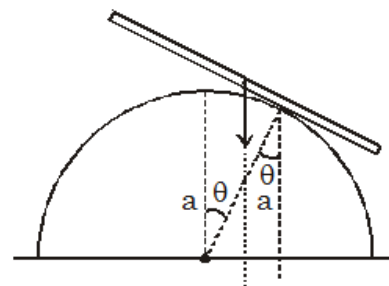
$$T = \mu(r\omega)^2$$

$$T = (0.1)(100)^2$$

$$T = 10^3 \text{ N}$$

$$V = \sqrt{\frac{10^3}{0.1}} = 100 \text{ m/s}$$

6. Ans (6.28)



θ is small

$$mga \sin \theta = \frac{m(2l)^2}{12} \theta$$

$$mga \theta = \frac{m\ell^2}{3} \alpha$$

$$\alpha = \left(\frac{3ga}{\ell^2}\right) \theta$$

$$T = \frac{2\pi}{\omega} = 2\pi \ell \sqrt{\frac{1}{3ga}}$$

$$\frac{T \sqrt{3ga}}{\ell} = 2\pi$$

7. **Ans (30.00)**

$$k = \frac{\pi}{3}$$

$$\omega = 10 \pi$$

$$v = \frac{\omega}{k} = 30 \text{ cm/s}$$

8. **Ans (1.57 to 1.58)**

$$T = \frac{T_1}{2} + \frac{T_2}{2}$$

$$T_1 = 2\pi\sqrt{\frac{L}{g}}, T_2 = 2\pi\sqrt{\frac{L}{3g}}$$

$$T = \pi\sqrt{\frac{L}{g}} + \pi\sqrt{\frac{L}{3g}}$$

$$\frac{T}{\pi\sqrt{\frac{L}{g}}} = 1 + \sqrt{\frac{1}{3}} = 1.58$$

CHEMISTRY

SEC-3

6. **Ans (4.00)**

1, 2, 3, 6

7. **Ans (5.00)**

1, 3, 6, 7, 8

8. **Ans (3.00)**

2, 5, 6

MATHEMATICS

SEC-1

1. **Ans (A,D)**

$$y = ax^2 + bx + c$$

$$-\frac{b}{2a} = 4; -\frac{(b^2 - 4ac)}{4a} = 2$$

$$\therefore b = -8a; b^2 - 4ac = -8a$$

$$\Rightarrow \therefore c = \frac{b^2 + 8a}{4a} \Rightarrow c = 2 + \frac{b^2}{4a}$$

$$\therefore c = 2 + \frac{64a^2}{4a} \Rightarrow c = 2 + 16a$$

$$\therefore \Delta = abc = a(-8a)(16a + 2)$$

$$\Rightarrow \Delta = -16a^2(8a + 1)$$

$$\Rightarrow \Delta = -16(8a^3 + a^2)$$

$$\Rightarrow \frac{d\Delta}{da} = -16(24a^2 + 2a)$$

which is -ve for $a \in [1, 3] \Rightarrow \Delta$ is $\downarrow$

$$\therefore \Delta_{\max} = -16(8(1)^3 + 1^2) = -144$$

$$\Delta_{\min} = -16(8(3)^3 + 3^2) = -3600$$

2. **Ans (A,B,C,D)**

$$S - S_n < \frac{1}{300}$$

$$\Rightarrow \frac{1}{1 - \frac{1}{3}} - \frac{1 - (\frac{1}{3})^n}{1 - \frac{1}{3}} < \frac{1}{300} \Rightarrow \frac{3}{2} \left(1 - 1 + \frac{1}{3^n} \right) < \frac{1}{300}$$

$$\Rightarrow \frac{1}{3^n} < \frac{1}{300} \times \frac{2}{3} \Rightarrow \frac{1}{3^n} < \frac{1}{450} \Rightarrow 3^n > 450$$

$\therefore$ Least $n = 6$.

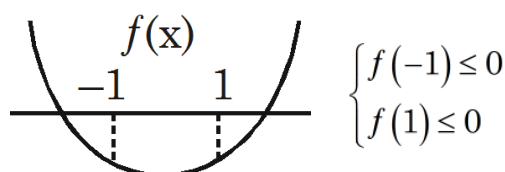
3. **Ans (A,B,D)**

$$f(x) = x^2 - (3k - 1)x + 2k^2 - 3k - 2 \geq 0$$

$$= (x - (k - 2))(x - (2k + 1))$$

$$x^2 - 1 \geq 0 \Rightarrow x \geq 1 \text{ or } x \leq -1$$

$\therefore$ for possible solutions



$$\Rightarrow \begin{cases} (-1 - k + 2)(-1 - 2k - 1) \leq 0 \\ (1 - k + 2)(1 - 2k - 1) \leq 0 \end{cases}$$

$$\Rightarrow \begin{cases} (k - 1)(k + 1) \leq 0 \\ (k)(k - 3) \leq 0 \end{cases} \Rightarrow k \in [0, 1]$$

4. **Ans (B,D)**

$$5^{x+2} > 5^{-2/x} \Rightarrow x + 2 > -\frac{2}{x}$$

$$\Rightarrow \frac{x^2 + 2x + 2}{x} > 0 \Rightarrow x > 0$$

5. **Ans (B)**

$$(\tan^{-1}x + \cot^{-1}x)^2 - 2(\tan^{-1}x) \left(\frac{\pi}{2} - \tan^{-1}x \right) = \frac{5\pi^2}{8}$$

$$\Rightarrow \frac{\pi^2}{4} - \pi \tan^{-1}x + 2(\tan^{-1}x)^2 = \frac{5\pi^2}{8}$$

$$\Rightarrow 2(\tan^{-1}x)^2 - \pi \tan^{-1}x - \frac{3\pi^2}{8} = 0$$

$$\Rightarrow \tan^{-1}x = -\frac{\pi}{4} \Rightarrow x = -1$$

6. **Ans (A,B,D)**

$$f(x) = \begin{cases} 4 & x \in \left(0, \frac{\pi}{2} \right) \\ -2 & x \in \left(\frac{\pi}{2}, \pi \right) \\ 0 & x \in \left(\pi, \frac{3\pi}{2} \right) \\ -2 & x \in \left(\frac{3\pi}{2}, 2\pi \right) \end{cases}$$

MATHEMATICS

SEC-2

1. **Ans (C)**

$$f(x) = (x-1)^4 (x-2)^3 (x-3)^2 (x-4)$$

Degree 10 polynomial

$\therefore$ Neither injective nor surjective

$$\text{Also, } f'''(1) = f'''(2) = f'''(3) = 0$$

$$\text{and } f'(4) = 3(6)^3$$

2. **Ans (B)**

$$f(x) = (x-1)^4 (x-2)^3 (x-3)^2 (x-4)$$

Degree 10 polynomial

$\therefore$ Neither injective nor surjective

$$\text{Also, } f'''(1) = f'''(2) = f'''(3) = 0$$

$$\text{and } f'(4) = 3(6)^3$$

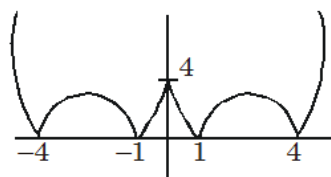
3. **Ans (D)**

$$x-1=t \Rightarrow x=t+1$$

$$\therefore f(t) = (t+1)^2 - 7(t+1) + 10$$

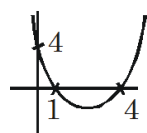
$$f(t) = t^2 - 5t + 4 \quad \forall t \in \mathbb{R}$$

Graph of $g(x)$



ND at 5 points.

Graph of $f(x)$:



$\therefore h(x)$ is DC at 2 points.

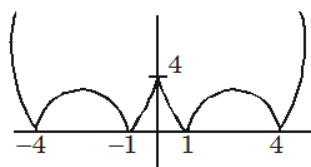
4. **Ans (B)**

$$x-1=t \Rightarrow x=t+1$$

$$\therefore f(t) = (t+1)^2 - 7(t+1) + 10$$

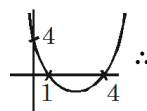
$$f(t) = t^2 - 5t + 4 \quad \forall t \in \mathbb{R}$$

Graph of $g(x)$



ND at 5 points.

Graph of $f(x)$:



$\therefore h(x)$ is DC at 2 points.

MATHEMATICS

SEC-3

1. **Ans (6.28)**

$$3^{\sec^2 x} \cdot \sqrt{y^2 - \frac{2y}{3} + \frac{2}{9}} \leq 1$$

$$\Rightarrow 3^{\sec^2 x} \cdot \sqrt{\left(y - \frac{1}{3}\right)^2 + \frac{1}{9}} \leq 1$$

$$\therefore 3^{\sec^2 x} \geq 3 \text{ and } \sqrt{\left(y - \frac{1}{3}\right)^2 + \frac{1}{9}} \geq \frac{1}{3}$$

$$\therefore \text{Only possibility : } \sec^2 x = 1; y = \frac{1}{3}$$

$$\Rightarrow x = 0, \pi, 2\pi, 3\pi$$

$$\therefore 6\pi \cdot \frac{1}{3} = 2\pi = 6.28$$

2. **Ans (1.00)**

On simplification, $f(x) = \cos 2x$

3. **Ans (1.00)**

$$\left(\frac{1 - 3\tan^2 1^\circ}{3 - \tan^2 1^\circ} \right) \cdot \frac{\cot 1^\circ}{\cot 3^\circ} = \left(\frac{1 - 3\tan^2 1^\circ}{3 \tan 1^\circ - \tan^3 1^\circ} \right) \cdot \frac{1}{\cot 1^\circ}$$

$$= \frac{1}{\tan 3^\circ \cot 3^\circ} = 1$$

4. **Ans (0.75)**

Put $x = 2, \frac{1}{2}, -1$ successively

$$f(2) + f\left(\frac{1}{2}\right) = 3$$

$$f\left(\frac{1}{2}\right) + f(-1) = \frac{3}{2}$$

$$f(-1) + f(2) = 0$$

$$\text{solving above, } f(2) = \frac{3}{4}$$

5. **Ans (3.14)**

$$x^2 + (2 - \tan \theta)x - (1 + \tan \theta) = 0 \quad \begin{cases} \alpha \\ \beta \end{cases}$$

$$\alpha + \beta = \tan \theta - 2$$

$$\alpha\beta = -1 - \tan \theta$$

$$\Rightarrow \alpha + \beta + \alpha\beta = -3$$

$$\Rightarrow (\alpha + 1)(\beta + 1) = -2$$

$$\therefore \text{C I : } \alpha + 1 = 2; \beta + 1 = -1$$

$$\Rightarrow \alpha = 1; \beta = -2$$

$$\therefore \tan \theta = 1 \Rightarrow \theta = \frac{\pi}{4}$$

$$\text{C II : } \alpha + 1 = -2; \beta + 1 = 1$$

$$\Rightarrow \alpha = -3; \beta = 0$$

$$\therefore \tan \theta = -1 \Rightarrow \theta = \frac{3\pi}{4}$$

$$\therefore \text{sum} = \pi$$

6. **Ans (1.50)**

1^∞ form

$$\Rightarrow e^{\lim_{x \rightarrow 0} \frac{4}{5x^2} (4 - 4\cos^2 x + \alpha \sin \beta x^2)} = e^2$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{16\sin^2 x}{5x^2} + \frac{4\alpha \sin \beta x^2}{5x^2} = 2$$

$$\Rightarrow \frac{4\alpha\beta}{5} + \frac{16}{5} = 2 \Rightarrow \alpha\beta = -\frac{3}{2}$$

7. **Ans (3.00)**

$$g'(f(x)) = \frac{1}{f'(x)} \quad \dots(1)$$

$$f(x) = x^3 + e^{x/2} \Rightarrow f'(x) = 3x^2 + \frac{e^{x/2}}{2}$$

$$f(x) = 1 + \sqrt{e} \text{ at } x = 1$$

$\therefore$ put $x = 1$ in (1) :

$$g'(1 + \sqrt{e}) = \frac{1}{3 + \frac{\sqrt{e}}{2}} = \frac{2}{6 + \sqrt{e}}$$

8. **Ans (3.00)**

$$L = - \left(\lim_{x \rightarrow \infty} x \ln x + 2x \ln \sin \left(\frac{1}{\sqrt{x}} \right) \right)$$

$$\text{Put } x = \frac{1}{t^2}$$

$$\therefore L = - \lim_{t \rightarrow 0} \frac{2 (\ln \sin t - \ln t)}{t^2}$$

$$\Rightarrow L = - \lim_{t \rightarrow 0} 2 \left[\frac{\cot t - \frac{1}{t}}{2t} \right] \text{ (Using L.H.)}$$

Rule)

$$\Rightarrow L = -2 \lim_{t \rightarrow 0} \frac{t - \tan t}{2t^3}$$

$$\Rightarrow L = \frac{1}{3}$$